MIL-M-3685B
26 December 1978
SUPERSEDING
MIL-M-3685A
19 March 1964

MILITARY SPECIFICATION

MACHINES, COMPUTING: DESK TYPE, AND PORTABLE TYPE WITH CARRYING CASE: PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements for the preservation, packing and marking of machines, computing, desk type and portable type with carrying case, motor driven and hand operated for shipment and storage.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-L-800	-	Lubricating Oil, General Purpose Preservative (Water-displacing, Low Temperature).
PPP-B-585	-	Boxes; Wood, Wirebound.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC PACK

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PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods Of.

STANDARDS

FEDERAL

FED-STD-101 - Preservation, Packaging and Packing Materials: Test Procedures.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials used for cleaning, preserving, packing and marking shall conform to the applicable specification as specified herein (see 6.5).

3.2 Disassembly. The extent of disassembly shall be determined by the manufacturer. This determination shall be based on the need for protection for fragile or projecting parts and for the reduction of cubage. Parts removed shall be adequately identified with tags. Where disassembly affects the operation of a computing machine, warning tags or labels indicating the disassembly shall be attached. The instructions for disassembly shall be attached. The instructions for assembling shall be included either as a part of the warning tag or label or as a separate sheet or folder attached to the computing machines. The disassembly shall not alter calibrations or settings which affect desired over-all operation of the computing machine and shall not be such that special skills or special tools or both are required to place the computing machine in operation.

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* 3.3 Preservation. Preservation shall be either Option 1 or Option II at the discretion of the contractor. The level of preservation shall be level A or Commercial as specified (see 6.2).

* 3.3.1 Level A. Each machine shall be preserved and unit packed Method IIa or IIb of MIL-P-116, except the humidity indicator shall not be required. Cleaning shall be accomplished by Method C-1 of MIL-P-116. When Method IIb is used, the interior container shall conform to style optional, type CF, variety SW or DW, grade 200 minimum of PPP-B-636, and the outer container shall be as specified in 3.4.1.1 or 3.4.1.2 for the applicable level of pack. The machine shall be thoroughly dried after cleaning, and preservative oil conforming to VV-L-800 shall be applied to those metal surfaces on which corrosion would inpair the usefulness of the machine.

* 3.3.1.1 Option I. For Option I, the rough handling test specified in MIL-P-116 shall be required only when specified (see 6.2).

* 3.3.1.2 Option II. For Option II, the unit packing shall be accomplished in a manner that will enable the packed machine to meet the test requirements as specified in 4.3.

* 3.3.2 Commercial. Each machine shall be preserved in accordance with MIL-STD-1188.

* 3.4 Packing. Packing shall be either Option I or Option II at the choice of the contractor. The level of packing shall be level A, B, or Commercial, as specified (see 6.2).

3.4.1 Option I packing.

* 3.4.1.1 Level A packing. Each machine, preserved as specified in 3.3, shall be individually packed in a shipping container conforming to overseas type, style A or B, type 2 load of PPP-B-601 or class 2, style 4, type 2 load of PPP-B-621. Each box containing a Method IIa pack shall be provided with a fiberboard liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Each shipping container shall be closed and reinforced in accordance with the applicable container specification.

* 3.4.1.2 Level B packing. Each machine, preserved as specified in 3.3, shall be individually packed in a shipping container conforming to style RSC, V3c, V3s, or V4s of PPP-B-636; class 2, style E of PPP-B-640; or domestic type, style A or B, type 2 load of PPP-B-601. Each shipping container shall be closed in accordance with the applicable container specification.

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* 3.4.1.3 Commercial packing. Each machine, preserved as specified in 3.3, shall be packed in accordance with MIL-STD-1188.

* 3.4.2 Option II. The choice of design and type shipping container and the packing of the machine is left to the option of the contractor, however, the choice of a shipping container for level A packing is limited to the selection of containers listed in 3.4.2.1. The completely packed container shall be capable of passing the tests as specified in section 4 for levels A and B shipments.

* 3.4.2.1 Level A (Option II). Each machine preserved as specified in 3.3, shall be packed in a manner that shall meet the test requirements for vibration, impact or drop, and compression for level A shipments as specified in 4.3. The pack shall have satisfactorily passed the tests, if upon examination the product is free from damage and the container and inner packing have only superficial scuffs, scratches or dents. The machine shall be packed in a shipping container constructed, closed, and strapped in conformance with one of the following specifications, as applicable.

PPP-B-585, class 3
PPP-B-601, overseas type
PPP-B-621, class 2

Inner packing materials shall be of water-resistant types.

3.4.2.2 Level B (Option II). Each machine, preserved as specified in 3.3, shall be packed in a manner that shall meet the test requirements for vibration, impact or drop, and compression for level B as specified in 4.3. The pack shall have satisfactorily passed the tests, if upon examination, the product is free from damage and the container and inner packing have only superficial scuffs, scratches or dents. Shipping containers and packing materials may be those normally used by the contractor for shipment of the machines provided the complete pack is capable of passing the test requirements of 4.3. Fiberboard shipping containers, when used, shall be of the weather-resistant type.

* 3.4.2.3 Commercial (Option II). Commercial packing for Option II shall be as specified in 3.4.1.3.

* 3.5 Marking. In addition to any special marking required by the contract, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

* 3.6 Workmanship. The preservation, packing and marking methods specified herein shall be uniformly applied to permit acceptance of the complete packaging requirements.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance. When certificates of compliance are required (see 6.2), the Government reserves the right to check test the items for compliance.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.2.2 Examination of preservation. Examination of preservation and marking of the interior packing to determine compliance with 3.3.1.1 and 3.5, shall be in accordance with MIL-P-116, as applicable.

4.2.3 Inspection of the end item.

4.2.3.1 Examination of end item. When shipping containers are required to be in accordance with one of the specifications specified in 3.4.1, the examination for closure and strapping shall be in accordance with the Quality Assurance Provisions contained in those specifications. Except the following defect shall be included: Marking omitted, incomplete, incorrect, illegible, or not in accordance with contract requirements.

4.2.4 Testing of end item. After all of the specified tests have been accomplished, a visual examination of the shipping container and its contents shall be made. Any damage to the computing machine shall be cause for rejection, and any damage to the shipping container or packing and packaging materials, except for superficial scuffs, scratches, or dents shall be cause for rejection.

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4.3 Tests. For tests specified herein, packed machines are classified into two gross weight groups as follows (see 6.4):

Weight-test group 1 - Machines weighing 100 pounds (45.4 kg) or more
Weight-test group 1A - Machines weighing less than 100 pounds (45.4 kg)

4.3.1 Weight-test group 1. One specimen weighing 100 pounds (45.4 kg) or more shall be subjected to a cycle of tests in the order herein specified:

* 4.3.1.1 Vibration test. The vibration test shall be conducted in accordance with Method 5019 of FED-STD-101, except the duration of tests shall be a minimum of 2 hours for level A or B.

4.3.1.2 Impact (longitudinal shock) test. Impact test shall be conducted in accordance with Method 5023 of FED-STD-101, except that a dummy load shall not be allowed for the impact test of the specimen. A dummy load may be utilized to predetermine the dolly release position on the incline-impact tester (see 6.3). The velocity shall be 7 feet (2.1 m) per second. Level A packs shall be subject to 2 test cycles and level B packs shall be subject to one test cycle.

4.3.1.3 Compression test. Compression test shall be conducted in accordance with Method 5016 of FED-STD-101, except that a dummy load shall not be allowed and the S factor shall be 4.00 for level A and 3.00 for level B.

4.3.2 Weight-test group 1A. One specimen weighing less than 100 pounds (45.4 kg) shall be subjected to a cycle of tests in the order specified as follows:

4.3.2.1 Vibration test. The vibration tests shall be performed in accordance with 4.3.1.1.

* 4.3.2.2 Drop test. The drop test shall be conducted in accordance with Method 5007 of FED-STD-101, except that the packed computing machine shall be dropped from the prescribed height in the following sequence:

1. A corner drop on the corner formed by the rear end, top, and right side 5-2-1 (see figure 1).
2. An edge drop on the shortest edge radiating from that corner.
3. An edge drop on the next shortest edge radiating from that corner.
4. An edge drop on the longest edge radiating from that corner.
5. A flatwise drop on one of the smallest faces.
6. A flatwise drop on the opposite smallest face.
7. A flatwise drop on one of the medium faces.
8. A flatwise drop on the opposite face.
9. A flatwise drop on one of the largest faces.
10. A flatwise drop on the opposite largest face.

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4.3.2.3 Compression test. The compression test shall be in accordance with 4.3.1.3.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

6.1 Intended use. The preservation and packing requirements of this specification are for the preparation of machines covered by this specification for delivery as follows:

Option I - Detailed and specific requirements for preservation and packing of computing machines
Option II - Preshipment performance type requirements for preservation and packing of computing machines

6.2 Ordering data. Requests, requisitions, schedules and contracts or orders should state the following:

(a) Title, number and date of this specification.
(b) Level of preservation (see 3.3).
(c) Level of packing (see 3.4).
(d) Whether rough handling test is required for preservation (see 3.3.1.1).
(e) Whether a certificate of compliance is acceptable (see 4.1.1).

6.3 Instrumentation source. A source of supply for a recorder that may be used in performing the impact test described in 4.3.1.2 is:

Impact Register Company, Champaign, IL

* 6.4 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

* 6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

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6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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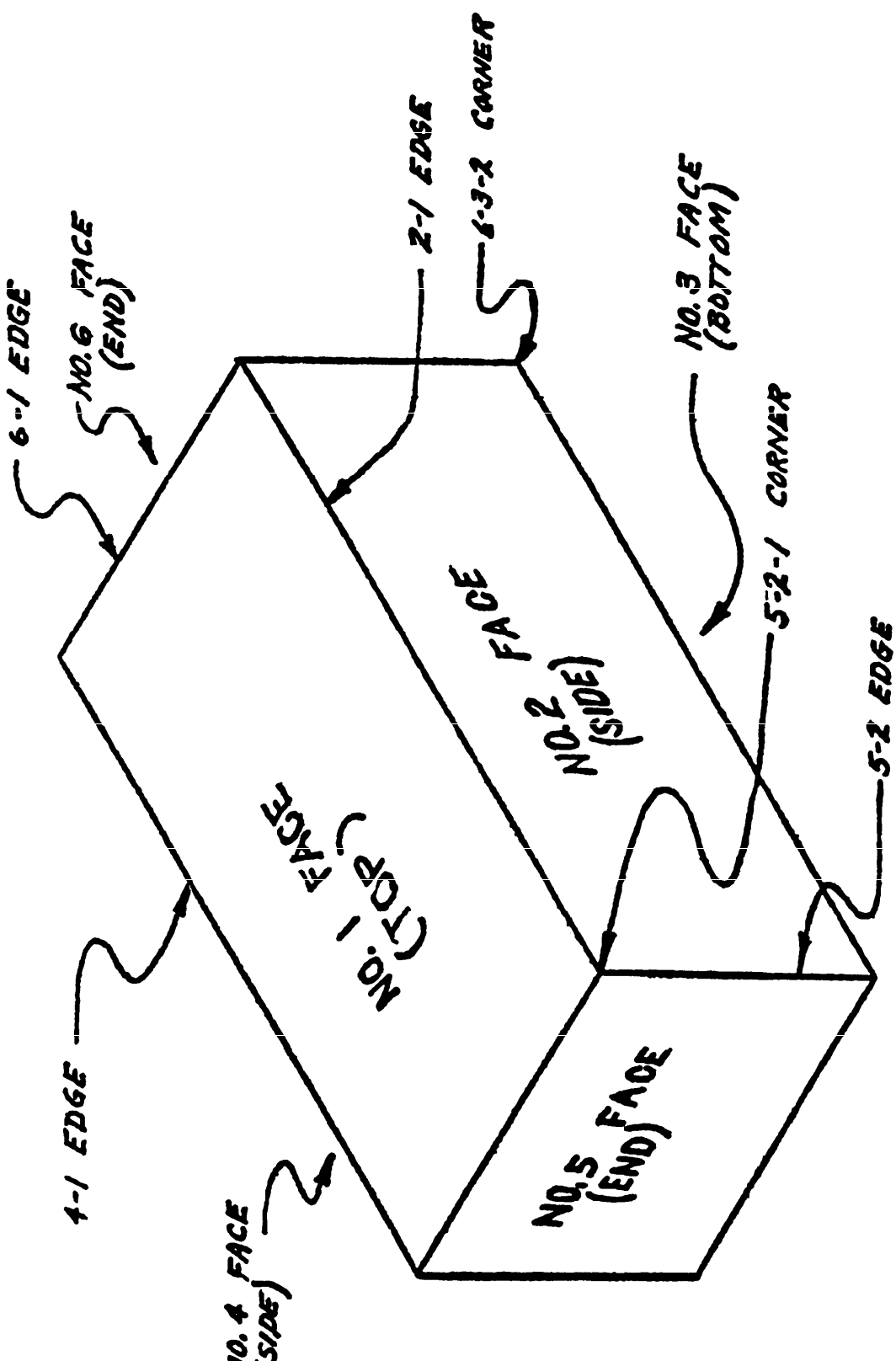


FIGURE 1. *Typical identification of container faces, edges and corners.*

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